

TraderJoes

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0.1 CISD 43- Trader Joe's - REDDIT

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0.1.1 Intro.

Scraping the subreddit 'r/traderjoes and performing a text analysis

0.1.2 Installation of Required Libraries

```
[4]: ! pip install emoji  
! pip install textblob  
! pip install wordcloud  
! pip install nltk
```

```
Requirement already satisfied: requests in c:\users\tinap\anaconda3\lib\site-  
packages (2.32.2)  
Requirement already satisfied: charset-normalizer<4,>=2 in  
c:\users\tinap\anaconda3\lib\site-packages (from requests) (2.0.4)  
Requirement already satisfied: idna<4,>=2.5 in  
c:\users\tinap\anaconda3\lib\site-packages (from requests) (3.7)  
Requirement already satisfied: urllib3<3,>=1.21.1 in  
c:\users\tinap\anaconda3\lib\site-packages (from requests) (2.2.2)  
Requirement already satisfied: certifi>=2017.4.17 in  
c:\users\tinap\anaconda3\lib\site-packages (from requests) (2026.2.25)  
Requirement already satisfied: emoji in c:\users\tinap\anaconda3\lib\site-  
packages (2.15.0)  
Requirement already satisfied: textblob in c:\users\tinap\anaconda3\lib\site-  
packages (0.20.0)  
Requirement already satisfied: nltk>=3.9 in c:\users\tinap\anaconda3\lib\site-  
packages (from textblob) (3.9.4)  
Requirement already satisfied: click in c:\users\tinap\anaconda3\lib\site-  
packages (from nltk>=3.9->textblob) (8.1.7)  
Requirement already satisfied: joblib in c:\users\tinap\anaconda3\lib\site-  
packages (from nltk>=3.9->textblob) (1.4.2)  
Requirement already satisfied: regex>=2021.8.3 in  
c:\users\tinap\anaconda3\lib\site-packages (from nltk>=3.9->textblob)  
(2023.10.3)  
Requirement already satisfied: tqdm in c:\users\tinap\anaconda3\lib\site-
```

packages (from nltk>=3.9->textblob) (4.66.4)
 Requirement already satisfied: colorama in c:\users\tinap\anaconda3\lib\site-packages (from click->nltk>=3.9->textblob) (0.4.6)
 Requirement already satisfied: wordcloud in c:\users\tinap\anaconda3\lib\site-packages (1.9.6)
 Requirement already satisfied: numpy>=1.19.3 in c:\users\tinap\anaconda3\lib\site-packages (from wordcloud) (1.26.4)
 Requirement already satisfied: pillow in c:\users\tinap\anaconda3\lib\site-packages (from wordcloud) (10.3.0)
 Requirement already satisfied: matplotlib in c:\users\tinap\anaconda3\lib\site-packages (from wordcloud) (3.8.4)
 Requirement already satisfied: contourpy>=1.0.1 in c:\users\tinap\anaconda3\lib\site-packages (from matplotlib->wordcloud) (1.2.0)
 Requirement already satisfied: cycler>=0.10 in c:\users\tinap\anaconda3\lib\site-packages (from matplotlib->wordcloud) (0.11.0)
 Requirement already satisfied: fonttools>=4.22.0 in c:\users\tinap\anaconda3\lib\site-packages (from matplotlib->wordcloud) (4.51.0)
 Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\tinap\anaconda3\lib\site-packages (from matplotlib->wordcloud) (1.4.4)
 Requirement already satisfied: packaging>=20.0 in c:\users\tinap\anaconda3\lib\site-packages (from matplotlib->wordcloud) (25.0)
 Requirement already satisfied: pyparsing>=2.3.1 in c:\users\tinap\anaconda3\lib\site-packages (from matplotlib->wordcloud) (3.0.9)
 Requirement already satisfied: python-dateutil>=2.7 in c:\users\tinap\anaconda3\lib\site-packages (from matplotlib->wordcloud) (2.9.0.post0)
 Requirement already satisfied: six>=1.5 in c:\users\tinap\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib->wordcloud) (1.16.0)
 Requirement already satisfied: nltk in c:\users\tinap\anaconda3\lib\site-packages (3.9.4)
 Requirement already satisfied: click in c:\users\tinap\anaconda3\lib\site-packages (from nltk) (8.1.7)
 Requirement already satisfied: joblib in c:\users\tinap\anaconda3\lib\site-packages (from nltk) (1.4.2)
 Requirement already satisfied: regex>=2021.8.3 in c:\users\tinap\anaconda3\lib\site-packages (from nltk) (2023.10.3)
 Requirement already satisfied: tqdm in c:\users\tinap\anaconda3\lib\site-packages (from nltk) (4.66.4)
 Requirement already satisfied: colorama in c:\users\tinap\anaconda3\lib\site-packages (from click->nltk) (0.4.6)

```

[162]: import time
import sys
import pandas as pd
import numpy as np
import json
import re
  
```

```
import emoji
from textblob import TextBlob
from wordcloud import WordCloud, STOPWORDS
import matplotlib.pyplot as plt

print("All imports successful!")
```

All imports successful!

0.1.3 Scraping with JSON

Access JSON file

```
[145]: ## default JSON Reddit caps results at 100 posts per request
subreddit = 'traderjoes'
limit = 100
timeframe = 'all' #hour, day, week, month, year, all
#listing = 'top' # controversial, best, hot, new, random, rising, top

def get_reddit(subreddit,listing,limit,timeframe):
    try:
        base_url = f'https://www.reddit.com/r/{subreddit}.json?
        limit={limit}&t={timeframe}'
        request = requests.get(base_url, headers = {'User-agent': '
        Practical-Ideal4175'})
    except:
        print('An Error Occured')
    return request.json()

tj_reddit = get_reddit(subreddit,listing,limit,timeframe)
```

0.1.4 Converting JSON to Dataframe

```
[148]: df = pd.json_normalize(tj_reddit["data"]["children"])
print(df.shape)
df.head(5)
```

(100, 541)

```
[148]: kind data.approved_at_utc data.subreddit \
0    t3          None    traderjoes
1    t3          None    traderjoes
2    t3          None    traderjoes
3    t3          None    traderjoes
4    t3          None    traderjoes

                                data.selftext data.author_fullname \
0  The large tote bag is expected to be the same ...    t2_5jtaw2pc
1  For Pastel Mini Tote Bags - Many stores are es...    t2_5jtaw2pc
```

2	My boyfriend's review: "Holy F--- what are tho...	t2_q6ceemw
3	Has anyone tried the Meyer lemon cake mix?	t2_1idfs402kc
4	I went to my local Trader Joe's today after my...	t2_1a31siy529

	data.saved	data.mod_reason_title	data.gilded	data.clicked	\
0	False	None	0	False	
1	False	None	0	False	
2	False	None	0	False	
3	False	None	0	False	
4	False	None	0	False	

	data.title	...	\
0	Megathread: Trader Joe's Large Lavender Reusab...	...	
1	March 2026 Update: Trader Joe's Pastel Mini Ca...	...	
2	Cookies & Creme Cookie Mix (Review)	...	
3	Is the Meyer lemon cake mix good?	...	
4	One of the best days I've had in a while - Lefse	...	

	data.media_metadata.bz1x9rngaxg1.s.u	data.media_metadata.bz1x9rngaxg1.id	\
0	NaN	NaN	
1	NaN	NaN	
2	NaN	NaN	
3	NaN	NaN	
4	NaN	NaN	

	data.media_metadata.t4lhorngaxg1.status	\
0	NaN	
1	NaN	
2	NaN	
3	NaN	
4	NaN	

	data.media_metadata.t4lhorngaxg1.e	data.media_metadata.t4lhorngaxg1.m	\
0	NaN	NaN	
1	NaN	NaN	
2	NaN	NaN	
3	NaN	NaN	
4	NaN	NaN	

	data.media_metadata.t4lhorngaxg1.p	data.media_metadata.t4lhorngaxg1.s.y	\
0	NaN	NaN	
1	NaN	NaN	
2	NaN	NaN	
3	NaN	NaN	
4	NaN	NaN	

	data.media_metadata.t4lhorngaxg1.s.x	\
--	--------------------------------------	---

0	NaN
1	NaN
2	NaN
3	NaN
4	NaN

	data.media_metadata.t4lhornga0xg1.s.u	data.media_metadata.t4lhornga0xg1.id
0	NaN	NaN
1	NaN	NaN
2	NaN	NaN
3	NaN	NaN
4	NaN	NaN

[5 rows x 541 columns]

Exploring/Cleaning data

```
[151]: df.columns
```

```
[151]: Index(['kind', 'data.approved_at_utc', 'data.subreddit', 'data.selftext',
            'data.author_fullname', 'data.saved', 'data.mod_reason_title',
            'data.gilded', 'data.clicked', 'data.title',
            ...,
            'data.media_metadata.bz1x9rnga0xg1.s.u',
            'data.media_metadata.bz1x9rnga0xg1.id',
            'data.media_metadata.t4lhornga0xg1.status',
            'data.media_metadata.t4lhornga0xg1.e',
            'data.media_metadata.t4lhornga0xg1.m',
            'data.media_metadata.t4lhornga0xg1.p',
            'data.media_metadata.t4lhornga0xg1.s.y',
            'data.media_metadata.t4lhornga0xg1.s.x',
            'data.media_metadata.t4lhornga0xg1.s.u',
            'data.media_metadata.t4lhornga0xg1.id'],
            dtype='object', length=541)
```

Cleaning up the data

```
[102]: # Selecting only the relevant columns
cols = [
    "data.title",
    "data.author",
    "data.subreddit",
    "data.upvote_ratio",
    "data.total_awards_received",
    "data.permalink",
    "data.created_utc",
    "data.selftext"
]
```

```

df_clean = df[cols]

df_clean.columns = [c.replace("data.", "") for c in df_clean.columns]

# Converting the timestamp to a readable date
from datetime import datetime
df_clean["created_utc"] = pd.to_datetime(df_clean["created_utc"], unit="s")

df_clean = df_clean.sort_values("created_utc", ascending=False).
    ↪reset_index(drop=True)
print(df_clean.shape)
df_clean.head(10)

```

(100, 8)

C:\Users\tinap\AppData\Local\Temp\ipykernel_23120\3120131003.py:20:

SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
df_clean["created_utc"] = pd.to_datetime(df_clean["created_utc"], unit="s")
```

```

[102]:

```

	title	author \
0	Synergistically seasoned popcorn	makooba
1	Review: Costa Rica Coffee - small lot	BlueKayak
2	Cookies & Creme Cookie Mix (Review)	bunnyfloofington
3	Is the Meyer lemon cake mix good?	NoCaterpillar800
4	One of the best days I've had in a while - Lefse	Northern_Lights_2
5	Sea salt chocolate chunk cookies	Churlishbeast
6	Taco Seasoned Gouda Cheese is incredible!!	lankaxhandle
7	Trader Joe's dupes and skincare: hair oil, bod...	Just_Art_6537
8	Spicy Spuds turned into Cheesy Fiesta Potatoes	Old_Bottle_2453
9	Need to make small emergency prom bouquets/nos...	beockwaymetcalf

	subreddit	upvote_ratio	total_awards_received \
0	traderjoes	0.95	0
1	traderjoes	0.92	0
2	traderjoes	0.98	0
3	traderjoes	0.98	0
4	traderjoes	0.98	0
5	traderjoes	0.55	0
6	traderjoes	0.93	0
7	traderjoes	0.82	0
8	traderjoes	0.97	0
9	traderjoes	0.50	0

```

                                permalink      created_utc  \
0  /r/traderjoes/comments/1t23gej/synergistically... 2026-05-02 22:13:31
1  /r/traderjoes/comments/1t23d0d/review_costa_ri... 2026-05-02 22:09:23
2  /r/traderjoes/comments/1t20131/cookies_creme_c... 2026-05-02 19:56:43
3  /r/traderjoes/comments/1t1xm6u/is_the_meyer_le... 2026-05-02 18:24:58
4  /r/traderjoes/comments/1t1fpfl/one_of_the_best... 2026-05-02 04:24:10
5  /r/traderjoes/comments/1t1eiab/sea_salt_chocol... 2026-05-02 03:23:54
6  /r/traderjoes/comments/1t1e49u/taco_seasoned_g... 2026-05-02 03:05:10
7  /r/traderjoes/comments/1t1dt2z/trader_joes_dup... 2026-05-02 02:50:32
8  /r/traderjoes/comments/1t1da0j/spicy_spuds_tur... 2026-05-02 02:24:58
9  /r/traderjoes/comments/1t1c99p/need_to_make_sm... 2026-05-02 01:36:51

```

```

                                selftext
0  Amazing! Perfectly described on the bag. 160 c...
1  [https://www.traderjoes.com/home/products/pdp/...
2  My boyfriend's review: "Holy F--- what are tho...
3      Has anyone tried the Meyer lemon cake mix?
4  I went to my local Trader Joe's today after my...
5  I bought these not knowing what to expect but ...
6  This stuff is phenomenal! I ate a good chunk p...
7  I'm from Costa Rica and going to the states on...
8  I used the spicy spuds  to make makeshift che...
9  I just talked my kid and their pal into flower...

```

[]:

```

[154]: # replacing "" with nan
df_clean["selftext"] = df_clean["selftext"].replace("", np.nan)
# creating columns that fills empty selftext with the title text
df_clean["col_text"] = df_clean["selftext"].fillna(df_clean["title"])
# displaying differences
df_clean[["col_text", "title", 'selftext']]

```

```

[154]:                                col_text  \
0  Amazing! Perfectly described on the bag. 160 c...
1  [https://www.traderjoes.com/home/products/pdp/...
2  My boyfriend's review: "Holy F--- what are tho...
3      Has anyone tried the Meyer lemon cake mix?
4  I went to my local Trader Joe's today after my...
..
95      Organic Conversation Olive Trio
96  I'm so excited that the Jalapeño limeade is ba...
97  Confirmed at my local store that the savory sq...
98  The large tote bag is expected to be the same ...
99  For Pastel Mini Tote Bags - Many stores are es...

```

```

                                title \
0          Synergistically seasoned popcorn
1          Review: Costa Rica Coffee - small lot
2          Cookies & Creme Cookie Mix (Review)
3          Is the Meyer lemon cake mix good?
4  One of the best days I've had in a while - Lefse
..
                                ...
95         Organic Conversation Olive Trio
96         The organic Jalapeño limeade is back!
97         Savory Squares Returning end of month!!
98 Megathread: Trader Joe's Large Lavender Reusab...
99 March 2026 Update: Trader Joe's Pastel Mini Ca...

                                selftext
0  Amazing! Perfectly described on the bag. 160 c...
1  [https://www.traderjoes.com/home/products/pdp/...
2  My boyfriend's review: "Holy F--- what are tho...
3      Has anyone tried the Meyer lemon cake mix?
4  I went to my local Trader Joe's today after my...
..
                                ...
95                                         NaN
96  I'm so excited that the Jalapeño limeade is ba...
97  Confirmed at my local store that the savory sq...
98  The large tote bag is expected to be the same ...
99  For Pastel Mini Tote Bags - Many stores are es...

[100 rows x 3 columns]

```

[]:

Cleaning text data

```

[160]: def clean_text(text):
        """Clean transcript text by removing extra spaces and line breaks."""
        text = str(text)
        text = text.replace("\n", " ")
        text = re.sub(r"\s+", " ", text)
        text = emoji.demojize(text)
        return text.strip()

    if not df_clean.empty:
        df_clean["clean_text"] = df_clean["col_text"].apply(clean_text)
        display(df_clean[["col_text", "clean_text"]])

```

```

                                col_text \
0  Amazing! Perfectly described on the bag. 160 c...
1  [https://www.traderjoes.com/home/products/pdp/...

```



```

2 My boyfriend's review: "Holy F--- what are tho...
3     Has anyone tried the Meyer lemon cake mix?
4 I went to my local Trader Joe's today after my...
..
95 Organic Conversation Olive Trio
96 I'm so excited that the Jalapeño limeade is ba...
97 Confirmed at my local store that the savory sq...
98 The large tote bag is expected to be the same ...
99 For Pastel Mini Tote Bags - Many stores are es...

```

```

                                clean_text
0 Amazing! Perfectly described on the bag. 160 c...
1 [https://www.traderjoes.com/home/products/pdp/...
2 My boyfriend's review: "Holy F--- what are tho...
3     Has anyone tried the Meyer lemon cake mix?
4 I went to my local Trader Joe's today after my...
..
95 Organic Conversation Olive Trio
96 I'm so excited that the Jalapeño limeade is ba...
97 Confirmed at my local store that the savory sq...
98 The large tote bag is expected to be the same ...
99 For Pastel Mini Tote Bags - Many stores are es...

```

[100 rows x 2 columns]

0.2 Analyzing Sentiment

0.2.1 Polarity

```

[115]: def get_polarity(text):
        return TextBlob(text).sentiment.polarity

def get_subjectivity(text):
    return TextBlob(text).sentiment.subjectivity

if not df_clean.empty:
    df_clean["polarity"] = df_clean["clean_text"].apply(get_polarity)
    df_clean["subjectivity"] = df_clean["clean_text"].apply(get_subjectivity)

    display(df_clean[["clean_text", "polarity", "subjectivity"]].head())

```

```

                                clean_text  polarity  subjectivity
0 Amazing! Perfectly described on the bag. 160 c...  0.875000      0.950000
1 [https://www.traderjoes.com/home/products/pdp/...  0.019048      0.288095
2 My boyfriend's review: "Holy F--- what are tho...  0.750000      0.575000
3     Has anyone tried the Meyer lemon cake mix?  0.000000      0.000000

```

```
4 I went to my local Trader Joe's today after my... 0.069616 0.493101
```

0.2.2 Sentiment

```
[169]: def sentiment_label(score):
        if score > 0:
            return "Positive"
        elif score < 0:
            return "Negative"
        else:
            return "Neutral"

    if not df.empty:
        df_clean["sentiment"] = df_clean["polarity"].apply(sentiment_label)
        display(df_clean[["clean_text", "polarity", "sentiment"]].head())
```

```
                                clean_text  polarity sentiment
0  Amazing! Perfectly described on the bag. 160 c... 0.875000  Positive
1  [https://www.traderjoes.com/home/products/pdp/... 0.019048  Positive
2  My boyfriend's review: "Holy F--- what are tho... 0.750000  Positive
3      Has anyone tried the Meyer lemon cake mix? 0.000000  Neutral
4  I went to my local Trader Joe's today after my... 0.069616  Positive
```

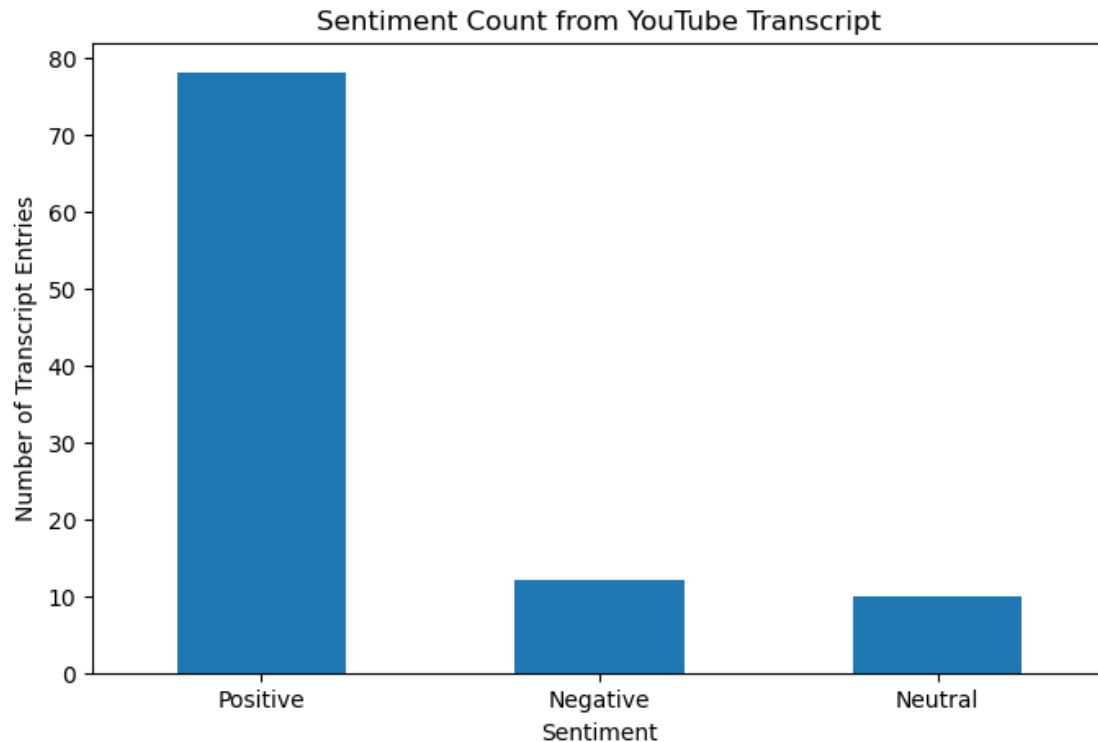
```
[171]: if not df_clean.empty:
        sentiment_counts = df_clean["sentiment"].value_counts().reset_index()
        sentiment_counts.columns = ["Sentiment", "Count"]
        display(sentiment_counts)
```

```
    Sentiment  Count
0  Positive      78
1  Negative      12
2  Neutral       10
```

0.2.3 Visualizations

bar chart

```
[174]: if not df_clean.empty:
        sentiment_counts.plot(kind="bar", x="Sentiment", y="Count", legend=False,
                                figsize=(8, 5))
        plt.title("Sentiment Count from YouTube Transcript")
        plt.xlabel("Sentiment")
        plt.ylabel("Number of Transcript Entries")
        plt.xticks(rotation=0)
        plt.show()
```



Pie Chart

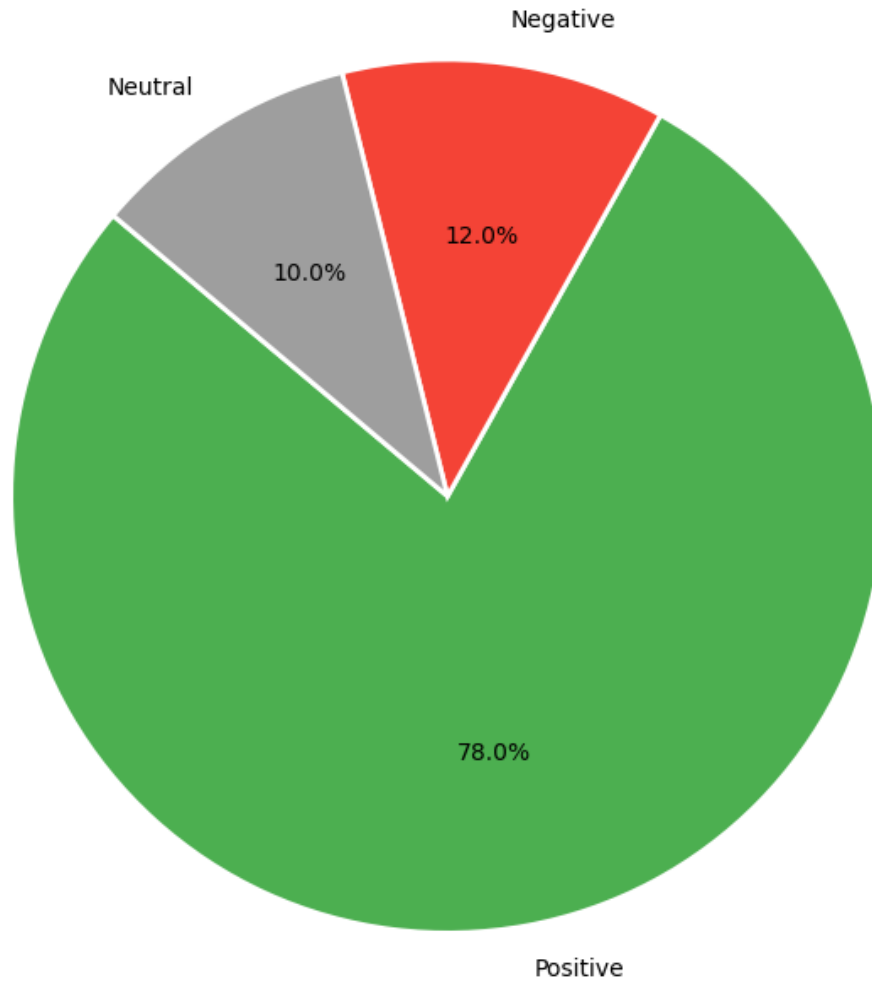
```
[178]: # Count sentiment categories
sentiment_counts = df_clean["sentiment"].value_counts()

# Define colors
colors = ["#4CAF50", "#F44336", "#9E9E9E"] # green, red, grey

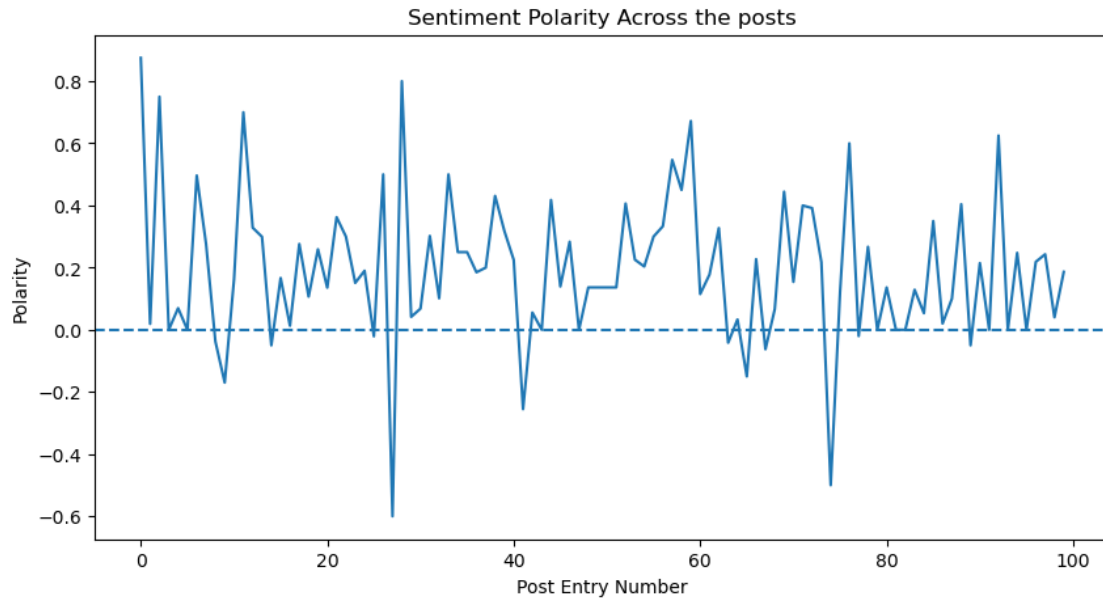
# Plot
plt.figure(figsize=(7, 7))
plt.pie(
    sentiment_counts,
    labels=sentiment_counts.index,
    colors=colors,
    autopct="%1.1f%%",      # show percentage on each slice
    startangle=140,         # rotate starting angle
    wedgeprops={"edgecolor": "white", "linewidth": 2}
)

plt.title("Trader Joe's Reddit Sentiment Analysis", fontsize=14,
          fontweight="bold")
plt.tight_layout()
plt.show()
```

Trader Joe's Reddit Sentiment Analysis



```
[125]: if not df_clean.empty:
plt.figure(figsize=(10, 5))
#plt.plot(df_clean.index, df_clean["polarity"])
plt.plot(df_clean.index, df_clean["polarity"])
plt.axhline(0, linestyle="--")
plt.title("Sentiment Polarity Across the posts")
plt.xlabel("Post Entry Number")
plt.ylabel("Polarity")
plt.show()
```



Word Blobs

```
[129]: import nltk
nltk.download('punkt_tab')
nltk.download('averaged_perceptron_tagger_eng') # ← updated name

from nltk.tokenize import word_tokenize
from nltk import pos_tag
```

```
[nltk_data] Downloading package punkt_tab to
[nltk_data] C:\Users\tinap\AppData\Roaming\nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data] C:\Users\tinap\AppData\Roaming\nltk_data...
[nltk_data] Package averaged_perceptron_tagger_eng is already up-to-
[nltk_data] date!
```

Word blobs of words in text that are tokenized as an adjective

```
[203]: full_text = " ".join(df_clean["clean_text"].dropna().tolist())

tokens = word_tokenize(full_text)
tagged_tokens = pos_tag(tokens)

vbz_list = [token for token, tag in tagged_tokens if tag in ['JJ', 'JJR', 'JJZ', 'JJS', 'ADJ']]
```

Combine the token words into One Text

```
[206]: if vbz_list:
    full_text = " ".join(vbz_list)
    print("Total characters in transcript:", len(full_text))
    print()
    print(full_text[:500])
else:
    full_text = ""
    print("No transcript text is available.")
```

Total characters in transcript: 3374

cal [//www.traderjoes.com/home/products/pdp/costa-rica-coffee-081522 https
 //www.traderjoes.com/home/products/pdp/costa-rica-coffee-081522 dark other next
 hard delicious top Best boxed lemon local ' abysmal ' proper ' lovely more
 difficult such bright insane only cold ' much grown flat bread salty phenomenal
 good ve good ve mixed ' ' ' ' t limited friendly good im fine cant sour better
 na mango pineapple next black other little good m burnt doesn favorite
 unexpected sprouted amazing i shredded

```
[210]: if vbz_list:
    custom_stopwords = set(STOPWORDS)

    wordcloud = WordCloud(
        width=900,
        height=450,
        background_color="white",
        stopwords=custom_stopwords,
        collocations=False
    ).generate(full_text)

    plt.figure(figsize=(12, 6))
    plt.imshow(wordcloud, interpolation="bilinear")
    plt.axis("off")
    plt.title("WordCloud of adjective words within the reddit posts for r/  

    ↪traderjoes")
    plt.show()
else:
    print("No transcript text is available for the WordCloud.")
```



Combine all posts into one text

```
[213]: if not df_clean.empty:
        full_text = " ".join(df_clean["clean_text"].dropna().tolist())

        print("Total characters in transcript:", len(full_text))
        print()
        print(full_text[:500])
    else:
        full_text = ""
        print("No transcript text is available.")
```

Total characters in transcript: 29524

Amazing! Perfectly described on the bag. 160 cal for 3 cups.

[<https://www.traderjoes.com/home/products/pdp/costa-rica-coffee-081522>] (<https://www.traderjoes.com/home/products/pdp/costa-rica-coffee-081522>) I saw these beans show up in our store (Northern Calif) about a week ago. The bag said "Medium roast" but with TJs it's a crapshoot on how dark they really are. I kept checking to see if any reviews on these beans had been posted here or other coffee forums, but no luck. Decided to bite the bull

```
[217]: if full_text.strip():
        custom_stopwords = set(STOPWORDS)

        wordcloud = WordCloud(
            width=900,
```


- Tutorial Sentiment Analysis with TextBlob
- youtube_MCM_Review

[]: